Legacy of redlining in current environmental (in)justice

Olivia Knapp

October 25, 2026

```
# Load packages
library(tidyverse)
```

```
── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
✔ dplyr     1.2.1     ✔ readr     2.2.0
✔ forcats   1.0.1     ✔ stringr   1.6.0
✔ ggplot2   4.0.3     ✔ tibble    3.3.1
✔ lubridate 1.9.5     ✔ tidyr     1.3.2
✔ purrr     1.2.2
── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()    masks stats::lag()
ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all
conflicts to become errors
```

```
library(tmap)
library(sf)
```

```
Linking to GEOS 3.12.1, GDAL 3.8.4, PROJ 9.4.0; sf_use_s2() is TRUE
```

```
library(spData)
library(knitr)
```

Historically red-lined neighborhoods in LA

```
#1. Create a map of historically red-lined neighborhoods, colored by HOLC grade
# Load data
la_map       <-       st_read('data/mapping-inequality/mapping-inequality-los-
angeles.json')
```

```
Reading layer `mapping-inequality-los-angeles' from data source
    `/home/oknapp/eds223/EDS223-HW2/data/mapping-inequality/mapping-inequality-
los-angeles.json'
  using driver `GeoJSON'
Simple feature collection with 417 features and 14 fields
```

```
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: -118.6104 ymin: 33.70563 xmax: -117.7028 ymax: 34.30388
Geodetic CRS:  WGS 84
```

```
la_ej                          <-                          st_read('data/ejscreen/
EJSCREEN_2023_BG_StatePct_with_AS_CNMI_GU_VI.gdb') |> filter(CNTY_NAME == 'Los
Angeles County')
```

```
Reading layer `EJSCREEN_StatePctiles_with_AS_CNMI_GU_VI' from data source
                                           `/home/oknapp/eds223/EDS223-HW2/data/
ejscreen/EJSCREEN_2023_BG_StatePct_with_AS_CNMI_GU_VI.gdb'
  using driver `OpenFileGDB'
Simple feature collection with 243021 features and 223 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: -19951910 ymin: -1617130 xmax: 16259830 ymax: 11554350
Projected CRS: WGS 84 / Pseudo-Mercator
```

```
crashes <- st_read('data/SWITRS_LA/Crashes.csv')
```

```
Reading layer `Crashes' from data source
  `/home/oknapp/eds223/EDS223-HW2/data/SWITRS_LA/Crashes.csv'
  using driver `CSV'
```

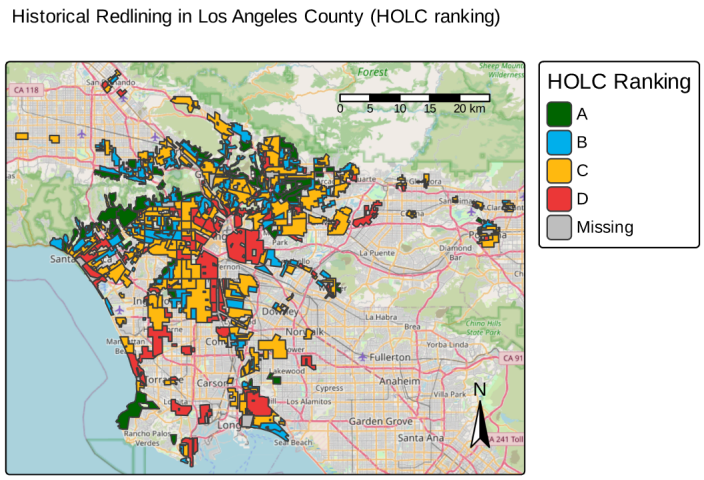
```
Warning: no simple feature geometries present: returning a data.frame or tbl_df
```

```
#Map red-lined areas in LA County
tm_basemap("OpenStreetMap") +
tm_shape(la_map) +
    tm_polygons(fill = 'grade',
                fill.scale  =  tm_scale(values  =  c('darkgreen',  'deepskyblue2',
'darkgoldenrod1', 'brown2')),
        fill.legend = tm_legend(title = "HOLC Ranking")) +
tm_scalebar(position = tm_pos_in("right", "top")) +
tm_compass(position = tm_pos_in('right', 'bottom')) +
tm_legend(position = tm_pos_out('right', 'center')) +
tm_layout(
    main.title = 'Historical Redlining in Los Angeles County (HOLC ranking)'
)
```

```
[v3->v4] `tm_layout()`: use `tm_title()` instead of `tm_layout(main.title = )`


── tmap v3 code detected ──────────────────────────────────────────────────────

[v3->v4] `tm_legend()`: use 'tm_legend()' inside a layer function, e.g.
'tm_polygons(..., fill.legend = tm_legend())'
[plot mode] fit legend/component: Some legend items or map compoments do not
fit well, and are therefore rescaled.
ℹ Set the tmap option `component.autoscale = FALSE` to disable rescaling.
```




```
# Summary table of % census block groups within and outide of each HOLC grade

# Join holc data and ej data
# Check if coordinate reference systems match
if(st_crs(la_map) == st_crs(la_ej)){
  print('File coordinate reference systems match')
} else{
  warning('File coordinate reference systems do not match.  Transform one file
before joining')
}
```

```
Warning: File coordinate reference systems do not match.  Transform one file
before joining
```

```
la_map <- st_make_valid(la_map)
```

```
# Transform la_ej to match la_mpa
ej_transformed <- st_transform(la_ej, crs = st_crs(la_map))

# Confirm transformation
if(st_crs(la_map) == st_crs(la_ej)){
  print('File coordinate reference systems match')
} else{
  warning('File coordinate reference systems do not match.  Transform one file
before joining')
}
```

```
Warning: File coordinate reference systems do not match.  Transform one file
before joining
```

```
# Check whether census blocks overlap
if (any(st_intersects(la_map, ej_transformed, sparse = FALSE))){
  print('census blocks overlap')
} else {
  print("census blocks do not overlap")
}
```

```
[1] "census blocks overlap"
```

```
#Convert to planar data
sf_use_s2(FALSE)
```

```
Spherical geometry (s2) switched off
```

```
# Join files
holc_ej <- st_join(y = la_map, x = ej_transformed,  join = st_intersects, left
= TRUE) |> st_drop_geometry()
```

```
although coordinates are longitude/latitude, st_intersects assumes that they
are planar
```

```
#Confirm join success
if (nrow(holc_ej) == nrow(ej_transformed)){
  print('Join was successful: all census blocks included')
```

```
} else {
  print("join failed.")
  print(paste('holc_ej has', nrow(holc_ej), 'rows'))
  print(paste('ej_transformed has ',nrow(ej_transformed), 'rows'))
}
```

```
[1] "join failed."
[1] "holc_ej has 8988 rows"
[1] "ej_transformed has  6591 rows"
```

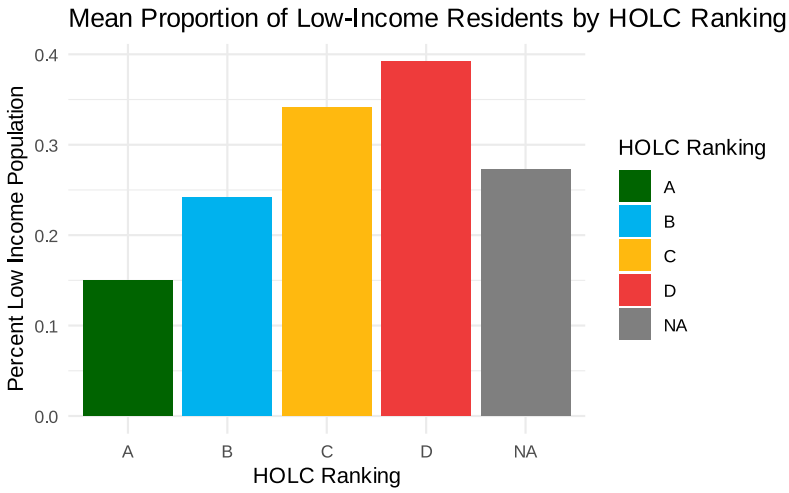
```
# 2. Create a table summarizing:
### percent of census blocks sithin each HOLC grade
### percent of cencus blocks that don't fall within a census grade

# Create the summarization table
holc_ej |> group_by(grade) |> summarize(
    Percent = ((n()/nrow(holc_ej))*100)
) |> knitr::kable(col.names = c("HOLC Grade", "Percent of Census Blocks"))
```

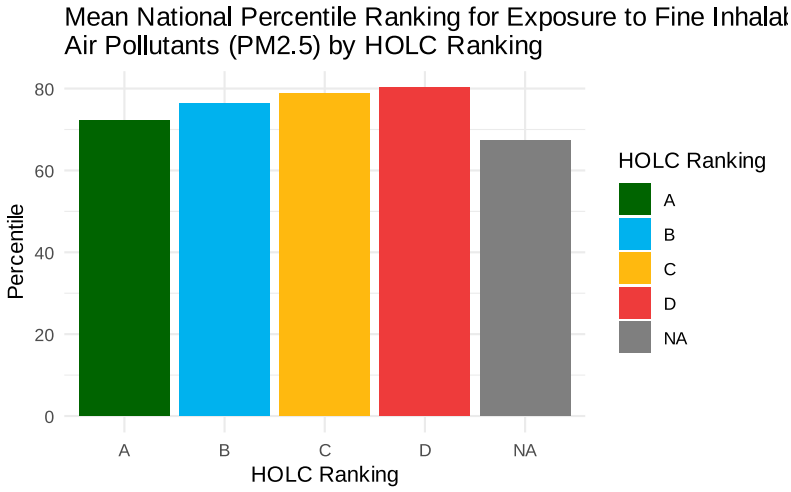
HOLC Grade	Percent of Census Blocks
A	4.99555
B	13.78505
C	34.02314
D	14.97552
NA	32.22074

Current Conditions by HOLC Ranking

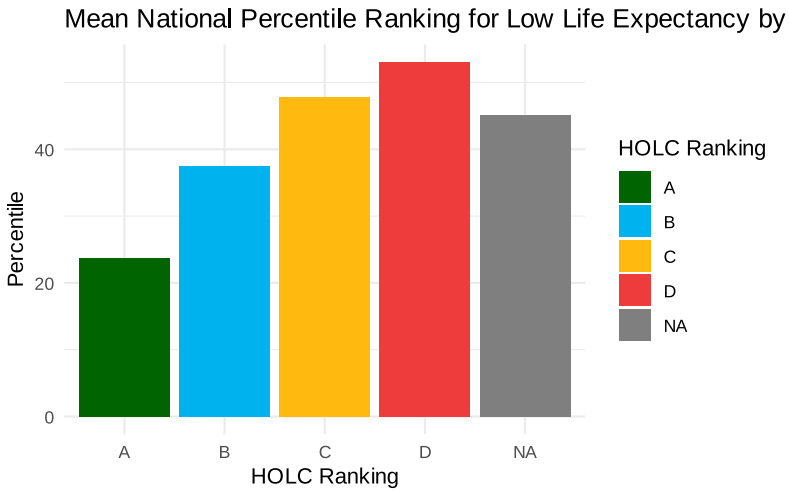
```
# Plot percent Low Income by HOLC Ranking
holc_ej  |>  group_by(grade)  |>  drop_na(LOWINCPCT)|>  summarize(low_inc_pct  =
mean(LOWINCPCT)) |> ggplot(
    mapping  = aes(x = grade, y = low_inc_pct, fill = grade)) +
    geom_bar(stat = 'identity') +
    scale_fill_manual(values = c('darkgreen', 'deepskyblue2', 'darkgoldenrod1',
'brown2', 'lightgray'), name = 'HOLC Ranking') +
    xlab("HOLC Ranking") +
    ylab("Percent Low Income Population") +
    ggtitle("Mean Proportion of Low-Income Residents by HOLC Ranking") +
    theme_minimal()
```




```
# Plot percentile for PM2.5 by HOLC Ranking
holc_ej  |> group_by(grade)  |> drop_na(P_PM25)|> summarize(pm25_percentile =
mean(P_PM25, drop_na = TRUE)) |> ggplot(
    mapping  = aes(x = grade, y = pm25_percentile, fill = grade)) +
    geom_bar(stat = 'identity') +
    scale_fill_manual(values = c('darkgreen', 'deepskyblue2', 'darkgoldenrod1',
'brown2', 'lightgray'), name = 'HOLC Ranking') +
        xlab("HOLC Ranking") +
    ylab("Percentile") +
     ggtitle("Mean National Percentile Ranking for Exposure to Fine Inhalable
\nAir Pollutants (PM2.5) by HOLC Ranking") +
    theme_minimal()
```



```
# Plot percentile for Low Life Expectancy by HOLC Ranking
holc_ej       |>       group_by(grade)       |>       drop_na(P_LIFEEXPPCT)|>
summarize(lowlifeexp_percentile = mean(P_LIFEEXPPCT)) |> ggplot(
    mapping  = aes(x = grade, y = lowlifeexp_percentile, fill = grade)) +
    geom_bar(stat = 'identity') +
    scale_fill_manual(values = c('darkgreen', 'deepskyblue2', 'darkgoldenrod1',
'brown2', 'lightgray'), name = 'HOLC Ranking') +
    xlab("HOLC Ranking") +
    ylab("Percentile") +
     ggtitle("Mean National Percentile Ranking for Low Life Expectancy by HOLC
Ranking") +
    theme_minimal()
```



On average, census block proportions of people with low incomes, exposure to air pollutants, and life expectancy scale to HOLC Rankings. Areas receiving a ranking of A have the lowest mean proportion of people with low incomes, lowest mean percentile ranking for exposure to fine particulate air pollutants, and the lowest mean percentile ranking for low life expectancy. Areas receiving a D have the highest mean in all three metrics, with areas receiving scores of B and C having intermediate means. Nearly 15% of census blocks in Los Angeles county are located in areas that received a ranking of D. Those census blocks have a disporptionately higher population of low-income residents, at about 50% of their population. They rank in the 80th national percentile in exposure to fine inhabalble air particles and just over the 50th percentile for low life expectancy; they experience more air pollution of fine particles than 80% of census blocks across the US and lower life expectancy than 50%. In contrast, in areas that recieved a ranking of A (just under 5% of LA county's census blocks), less that 20% of the population is low-income. Areas that recieved an A still rank in the 70th percentile for fine air particle exposure, but just over the 20th percentile for life-expectancy. This indicates that the New Deal HOLC ranking system has resulted in measurable demographic differences decades later. Past diversion of investments from

areas with low rankings likely contributes to current disparities in wealth and environmental safety as redlined areas received less real estate investment and resulting social, safety, and environmental investment. Lower wealth and greater environmental health risks likely combine with other socioeconmic factors to create the observed disparity in life expectancy.

Part 2: Legacy of redlining in pedstrian and bike accidents

```
#Check NAs in Lattitude/Longitude vs Point_X/Point_Y
crashes |> filter(crashes['LONGITUDE'] == "NA") |> dim()
```

```
Warning: Using one column matrices in `filter()` or `filter_out()` was deprecated
in
dplyr 1.1.0.
ℹ Please use one dimensional logical vectors instead.
```

```
[1] 97350    80
```

```
# Missing data for 97350 rows
crashes |> filter(crashes['POINT_X'] == "NA") |> dim()
```

```
Warning: Using one column matrices in `filter()` or `filter_out()` was deprecated
in
dplyr 1.1.0.
ℹ Please use one dimensional logical vectors instead.
```

```
[1] 9484   80
```

```
# Missing data for 9484 rows
crashes_points <- crashes |> select(-c('LATITUDE', 'LONGITUDE'))

#Conver crash points to geometry class
crashes_spatial <- crashes_points |> mutate(
    POINT_X_numeric = as.numeric(POINT_X),
    POINT_Y_numeric = as.numeric(POINT_Y)
    )|> drop_na(
        POINT_Y_numeric, POINT_X_numeric
    ) |> st_as_sf(
        coords = c('POINT_X_numeric', 'POINT_Y_numeric'),
        crs = 4326) # Set crs to WGS 84
```

```
Warning: There were 2 warnings in `mutate()`.
The first warning was:
ℹ In argument: `POINT_X_numeric = as.numeric(POINT_X)`.
Caused by warning:
! NAs introduced by coercion
ℹ Run `dplyr::last_dplyr_warnings()` to see the 1 remaining warning.
```

```
#Check that CRS match
if(st_crs(crashes_spatial) == st_crs(la_map)){
  print('File coordinate reference systems match')
} else{
  warning('File coordinate reference systems do not match.  Transform one file
before joining')
}
```

```
[1] "File coordinate reference systems match"
```

```
# Transform crs of crashes points to match joined holc_ej
crashes_transformed <- st_transform(crashes_spatial, crs = st_crs(la_map))

#Confirm CRS change
if(st_crs(crashes_spatial) == st_crs(la_map)){
  print('File coordinate reference systems match')
} else{
  warning('File coordinate reference systems do not match.  Transform one file
before joining')
}
```

```
[1] "File coordinate reference systems match"
```

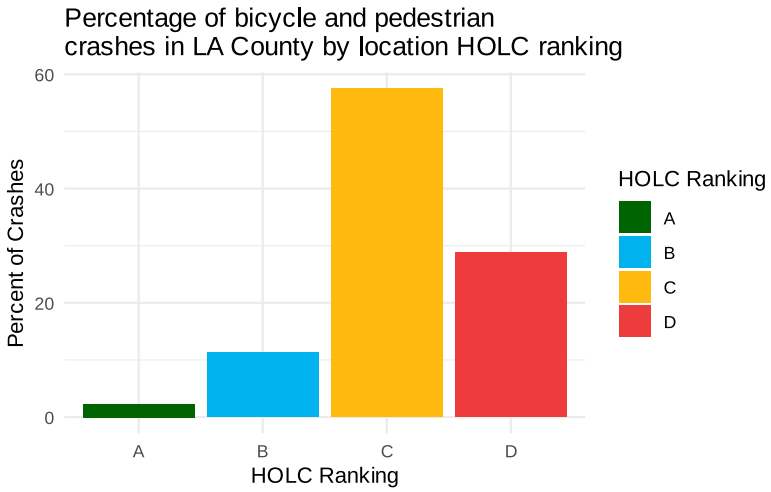
```
#Join with HOLC data
crashes_holc  <-  st_join(y  =  la_map,  x  =  crashes_transformed,    join  =
st_intersects, left = TRUE) |> st_drop_geometry()
```

```
although coordinates are longitude/latitude, st_intersects assumes that they
are planar
although coordinates are longitude/latitude, st_intersects assumes that they
are planar
```

```r
#Confirm join success
if (nrow(crashes_holc) == nrow(crashes_transformed)){
  print('Join was successful: all crashes included')
} else {
  print("join failed.")
  print(paste('crashes_holc has', nrow(crashes_holc), 'rows'))
  print(paste('crashes_transformed has ',nrow(crashes_transformed), 'rows'))
}
```

```
[1] "Join was successful: all crashes included"
```

```r
#Summarize bike/pedestrain crash trends by HOLC grade and plot by HOLC Ranking
crashes_bike_ped <- crashes_holc |> filter(BICYCLE_ACCIDENT == "Y" |
    PEDESTRIAN_ACCIDENT == "Y") |> drop_na(grade)
crashes_bike_ped |> group_by(grade
    ) |> summarize(percent = n()/nrow(crashes_bike_ped) *100
    ) |> ggplot(
      mapping  = aes(x = grade, y = percent, fill = grade)) +
    geom_bar(stat = 'identity') +
    scale_fill_manual(values = c('darkgreen', 'deepskyblue2', 'darkgoldenrod1',
'brown2', 'lightgray'),
      name = "HOLC Ranking") +
    xlab("HOLC Ranking") +
    ylab("Percent of Crashes") +
     ggtitle("Percentage of bicycle and pedestrian \ncrashes in LA County by
location HOLC ranking") +
    theme_minimal()
```



Pedestrian and cyclist safety appears to follow similar patterns with HOLC ranking as income and air pollution, though perentage of crashes peaks in areas that recieved a ranking of C rather than D. However, there is still a greater percentage of crashes in areas that received a D than A or B. This likely contributes to the disparity in life expectancy. Among areas that received an HOLC ranking, less than 20% of crashes occur in areas with a ranking of A or B, but nearly 60% occur in areas wth rankings of C and nearly 30% in areas ranked as D. This likely resulted from less investment in safety infrastucture and maintenance in these historically lower-income areas, especially if hazard reporting, funding, and county response times are influenced by income and demographic factors. To strategically improve pedestrian and cyclist safety, the County of LA can invest in more safety measures in historically redlined areas and critically examine how it makes decisions related to safety infrastructure investment and maintenance.

Revision Memo

To address the comment regarding appropriate spatial operations, I updated the join between the HOLC scores for LA neighborhoods (`la_map`) and data from EJScreen indicated by the comment in Part 1. I reveised it to join by `st_intersects` instead of `st_within` as the boundaries in each dataset are not the same. Though `st_intersects` does not necessarily need to be specified as it is the default argument for `st_join`, I assigned the argument explicity to emphasize the change. This resulted in changes to the graphs showing exisitng environmental injustice resulting from historical HOLC rankings, so I updated by interpretation accordingly.

I have resubmitted by GitHub repository link to the Google form and have also provided it here: https://github.com/gingersknapp/EDS223-HW2